Investigation into the effect of temperature on the equilibrium constant

RQ: How does a temperature change (10.0 ± 0.1°C, 20.0 ± 0.1°C, 30.0 ± 0.1°C, 40.0 ± 0.1°C, 50.0 ± 0.1°C) affect the equilibrium constant for the reversible reaction between 0.002 mol dm^{-3} iron(III) and 0.002 mol dm^{-3} thiocyanate?

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I. Introduction

1.1 Research question

How does a temperature change (10.0±0.1°C, 20.0±0.1°C, 30.0±0.1°C, 40.0±0.1°C, 50.0±0.1°C) affect the equilibrium constant for the reversible reaction between 0.002 mol dm^{-3} iron(III) and 0.002 mol dm^{-3} thiocyanate?

1.2 Introduction

During a chemistry demonstration, I observed the formation of the blood-red iron(III) thiocyanate complex. From textbooks and lectures, I was intrigued by the fact that the equilibrium constant (K_C), the ratio of products to reactants, is always constant despite concentration changes. However, there was only one factor that changes the equilibrium constant: temperature. This piqued my interest in investigating how thermal energy dictates the ratio of products to reactants in equilibrium. This investigation uses iron(III) nitrate nonahydrate and potassium thiocyanate, assuming instant equilibrium as temperature changes. A UV-Vis spectrophotometer measured absorbance at each temperature. Dividing absorbance by the molar extinction coefficient from a pre-made calibration curve gave the concentration, which is used to calculate K_C.

II. Background information

2.1 Background knowledge

A chemical reversible reaction of iron(III) and thiocyanate ions produces a blood-red complex:

$$Fe^{3+}_{(aq)} + SCN^{-}_{(aq)} \rightleftharpoons [Fe(SCN)]^{2+}_{(aq)}$$

This is a dynamic equilibrium where the concentrations of reactants and products remain constant once the forward (exothermic) and reverse (endothermic) reaction rates equalize. The resulting product is a coordinated compound where the thiocyanate ligands connect to the iron(III) center. The equilibrium constant is defined as:[1]

$$K_C = \frac{[[Fe(SCN)]^{2+}]_{eq}}{[Fe^{3+}]_{eq}[SCN^{-}]_{eq}}$$

The equilibrium constant (K_C) represents the specific ratio of products to reactants where a system is at its minimum state of free energy. This state is explained by Gibbs Free Energy (G).[1] A reaction occurs spontaneously as long as the Gibbs Free Energy of the system is decreasing ($\Delta G < 0$). When the system reaches the lowest possible point of its energy curve ($\Delta G = 0$), it cannot release energy anymore. At this

point, the system is at equilibrium. While ΔG indicates whether a reaction is spontaneous, the Standard Gibbs Free Energy change (ΔG°) informs where the equilibrium position is located.

$$\Delta G^\circ = -RT ln K_C$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

Where

ΔG°: Standard Gibbs Free energy change

ΔH°: Standard enthalpy change

ΔS°: Standard entropy change

T: Temperature(K).

ΔG° It is the fixed energy difference between pure reactants and pure products. If products are more stable ($\Delta G^\circ < 0$), the equilibrium shifts to the product side, resulting in a large K_C. While the position can alter with changes such as the concentration of species, the equilibrium constant does not. The only factor that influences the equilibrium constant at a position is temperature.

The effect of temperature on the equilibrium constant is explained qualitatively by Le Chatelier's principle, which states that a system tries to counteract the disturbance of a change in condition.[1] An increase in temperature favors the endothermic (reverse for this reaction) direction to absorb the excess thermal energy, thereby reducing the concentration of the red complex. Lowering the temperature also follows this principle.

It is also possible to understand the effect of temperature on the equilibrium constant quantitatively by deriving a mathematical equation from the two expressions of the Standard Gibbs Free Energy change (ΔG°). Combining the equations $\Delta G^\circ = -RT ln K_C$ and $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$, results in the following equation:[2]

$$ln K_C = -\frac{\Delta H^\circ}{R}\left(\frac{1}{T}\right) + \frac{\Delta S^\circ}{R}$$

This is the linear form of van't Hoff equation. According to it, the reciprocal of T is linearly proportional to the natural logarithm of K_C. It is assumed that ΔH° and ΔS° are constant as temperature changes, even

though they are temperature-dependent. Since we are investigating a small range of 10°C to 50°C, the assumption is valid.

Since the $[Fe(SCN)]^{2+}$ complex absorbs light specifically at a specific wavelength, Beer-Lambert's Law is used to determine the equilibrium concentration of product.[1]

$$A = \varepsilon c l$$

Where

A: Absorbance

ε The molar extinction coefficient ($dm^3mol^{-1}cm^{-1}$)

C: Molar concentration (mol dm^{-3})

L: Path length (cm).

2.2 Hypothesis

The forward reaction of $[Fe(SCN)]^{2+}$ is exothermic, so according to Le Chatelier's principle, the system will try to mitigate the rise in temperature by undergoing an endothermic reaction. Thus, as the temperature rises, the reverse reaction will occur, and the equilibrium constant will decrease.

III. Methodology

3.1 Justification of method

A spectrophotometer was chosen over titration for three reasons.

1. Real-time measurement: Titration is slow, and even with quenching, heat loss occurs before it finishes, while the spectrophotometer captures absorbance the instant the target temperature is reached.
2. Analytical sensitivity: it offers much lower uncertainty, below $\pm\ 0.0006$, than manual titration.
3. Sustainability: it avoids secondary titration reagents, minimizing chemical waste.

3.2 Scope of the research

The investigation is divided into two parts:

- Part A (Calibration): Creating a calibration curve (Concentration-Absorbance) to obtain the gradient for the molar extinction coefficient (ε), which serves as a key to calculate the concentration of $[Fe(SCN)]^{2+}$ with only its absorbance.
- Part B (Main research): Measuring the absorbance of $[Fe(SCN)]^{2+}$ at different temperatures (10.0 ±0.1°C, 20.0±0.1°C, 30.0±0.1°C, 40.0±0.1°C, 50.0±0.1°C) to calculate the equilibrium constant.

3.3 Part A: Calibration strategy and side reaction control

To ensure a known concentration of $[Fe(SCN)]^{2+}$ for standard solutions, one reactant must be in excess. According to Le Chatelier's principle, a large excess of one reactant ensures the limiting reagent reacts completely to form the complex ion.

For this experiment, Fe^{3+} has been chosen to be in excess. $[Fe(SCN)]^{2+}$ is a coordinate complex with iron(III) as the central ion with SCN^- as ligands. It primarily adopts an octahedral structure $[Fe(SCN)(H_2O)_5]^{2+}$ when SCN^- concentrations are low relative to Fe^{3+}. A large excess of SCN^- would fill all coordination sites on the Fe^{3+} ion, forming more intensely red colored red complexes such as $Fe(SCN)_3$ or $Fe(SCN)_4^-$.[3] This color change would create a systematic error when measuring the absorbance.

Another factor to consider is suppressing the hydrolysis of iron(III). Preliminary experiments showed that $Fe(NO_3)_3$ in pure water turned vivid yellow due to hydrolysis, rather than the expected pale yellow. This created huge spiky lines on the absorbance curve, which made it hard to read the measurements. Hydrolysis occurs because Fe^{3+} weakens the O-H bond in coordinated water:[4]

$$Fe^{3+}_{(aq)} + 3H_2O_{(l)} \rightleftharpoons Fe(OH)_{3(s)} + 3H^+_{(aq)}$$

The resulting $Fe(OH)_3$ precipitate alters the Fe^{3+} concentration and impacts absorbance readings. To prevent this, an HNO_3 was added to the solution. The high concentration of H^+ ions pushes the equilibrium to the left, reversing the formation of $Fe(OH)_3$ and bringing back Fe^{3+} ions.

$Fe(NO_3)_3$ and KSCN will be used to induce the reaction between iron and thiocyanate. The ions (Fe^{3+}, SCN^-), and spectator ions of K^+ and NO_3^- are pale yellow and colorless.[5] Therefore, the absorbance of blood red Iron(III) thiocyanate is obtained with minimal interference.

UV-Vis spectrophotometer parameters:

- Wavelength selection: The standard solution with the highest concentration was used to identify the peak absorbance wavelength, which was 458nm. This wavelength was kept constant for all measurements.
- Concentration adjustments: To keep the absorbance range below about 2 to minimize the spectrophotometer uncertainty, concentrations of $0.200_0 \pm 0.0005 moldm^{-3}$ Fe^{3+} and $0.0020 \pm 0.0001 moldm^{-3}$ SCN^- were used, determined through a preliminary experiment.

3.4 Part B: Main research

Preliminary trials, measuring the absorbance of the cooling sample, showed that the chemical reaction responds instantaneously to temperature changes. To manage heat loss from the transportation of the cuvette to the spectrophotometer after heating or cooling, the following procedure was adopted.

- High temperature trials (30.0±0.1°C, 40.0±0.1°C, 50.0±0.1°C): The sample was heated to 55.4± 0.1°C in a beaker of water on a hot plate. Absorbance was measured as the sample cooled. This accounts for heat loss during cuvette wiping and transfer.
- Room temperature trial (20.0±0.1°C): Measured at a room temperature of 19.7±0.1°C.
- Low temperature trial (10.0±0.1°C): The sample is cooled to 4.1±0.1°C in a beaker of ice water. Absorbance is recorded as the sample warms up to 10.0±0.1°C while inside the spectrophotometer.

The resulting absorbance values are divided by the molar extinction coefficient from part A to determine the concentration of $[Fe(SCN)]^{2+}$ and calculate the equilibrium constant.

3.5 Materials and Apparatus

Materials[6]		
Materials(formula)	**Characteristics**	**Wastewater treatment**
Iron(III) nitrate nonahydrate ($Fe(NO_3)_3 \cdot 9H_2O$)	-Molar mass: 404.00$gmol^{-1}$ -Usage: 4.102±0.001g -Purity: 98.5% -Deliquescent -Colour: Brownish orange -Melting point: 47.2°C -CAS No: 7782-61-8 -Strong oxidizing agent -Extremely acidic	Dispose of in a liquid waste container for acidic solutions.
Potassium thiocyanate (KSCN)	-Molar mass: 97.18$gmol^{-1}$ -Usage: 0.020±0.001g -Purity: 99.0% -Deliquescent	Dispose of in a liquid waste container for inorganic solutions

	-Colour: Colourless/White -Melting point: 173°C -CAS No: 333-20-0	
Nitric acid (HNO3)	-Density: 1.42gcm^{-3} -Usage: 3.410±0.004cm^3 -Purity: 65±1% -Colour: Colourless -Boiling point: 121°C -CAS No: 7697-37-2 -When exposed to heat, it decomposes into NO_2 -Strong acid	Dispose of in a liquid waste container for acidic solutions

Table 1: Materials used in the experiment

Apparatus (how many usedx capacity-name (±absolute uncertainty))	
Apparatus for part A	**Apparatus for part B**
12x 50cm^3-beakers 1x 50cm^3-volumetric flask (±0.12) 2x 100cm^3-volumetric flask (±0.20) 6x polystyrene cuvette 1x 1cm^3-micropipette (±0.001) 1x electronic scale (±0.001) 2x weighing paper 2x spatula 1x cuvette rack Pipette tip 1x disposable spoid 4AU-UV-visible spectrophotometer (Agilent8454 UV system)	1x digital thermometer (±0.1°C) 1x stand 1x clamp (to hold the thermometer) 1x hot plate 4x 50cm^3-beaker 1x polystyrene cuvette 1x 1cm^3-micropipette (±0.001) Pipette tip 1x 100cm^3-volumetric flask (±0.1) 1x disposable spoid 4AU-UV-visible spectrophotometer (Agilent8454 UV system)

Table 2: Apparatus used in the experiment

3.6 Variables

Independent Variable (IV): Temperature ($10.0\pm0.1°C$, $20.0\pm0.1°C$, $30.0\pm0.1°C$, $40.0\pm0.1°C$, $50.0\pm0.1°C$).

- Justification of range: A 10°C interval was chosen since 5°C increments showed insufficient absorbance change in preliminary trials. The upper limit was set at $50.0\pm0.1°C$ to prevent HNO_3 decomposition, which would alter concentrations.
- Method of control: Samples for $30.0\text{-}50.0\pm0.1°C$ trials will be overheated in a beaker of water on a hot plate and measured during cooling. The $20.0\pm0.1°C$ trial is measured at room temperature, and the $10.0\pm0.1°C$ trial involves pre-cooling in ice water, measured while warming.
- Measurement method: Since the UV-Vis spectrophotometer has a delay to measure the absorbance, the temperature when the button is pressed and when the measurement is done are both recorded.

Dependent Variable (DV): The equilibrium constant (K_C) of the reversible reaction $Fe^{3+} + SCN^- \rightleftharpoons [Fe(SCN)]^{2+}$

- Measurement method: Absorbance is measured via a UV-Vis spectrophotometer at 458nm. This absorbance is converted to concentration using the Part A calibration curve and Beer-Lambert's law to calculate K_C at each temperature.

Controlled Variables:

Controlled Variable	Explanation	Method of control
Wavelength for spectrophotometer readings	ε is wavelength specific. Different wavelengths per trial give inconsistent absorbance at the same concentration.	Fixed 458 nm (Part A peak) used for all readings.
Concentration of Nitric Acid (HNO_3)	HNO_3 suppresses Fe^{3+} hydrolysis. Thus, inconsistent acidity gives inconsistent suppression. Concentration above 0.8 mol dm^{-3} is hazardous.	Consistent $0.50_0\pm0.009$ mol dm^{-3} HNO_3 stock in every mixture, kept below 0.8 mol dm^{-3}.
Concentration/ Volume of Iron(III) Nitrate nonahydrate	Part A: Fe^{3+} excess ensures all SCN^- reacts (else invalidates known-concentration assumption).	Part A: $0.200_0\pm0.0005$ mol dm^{-3} to all stocks with volume $5.000\pm0.005cm^3$ using a

$(Fe(NO_3)_3 \cdot 9H_2O)$	Part B: similar volume/concentration of Fe^{3+} to that of SCN^- makes small temperature-driven shifts detectable.	micropipette. Part B: Stock solution was diluted to $0.00200_0 \pm 0.000009$ mol dm^{-3} with a volume of $4.000 \pm 0.004 cm^3$
Concentration/ Volume of Potassium Thiocyanate (KSCN)	Part A: SCN^- must be limited to fully react. Part B: Equal volume/concentration to Fe^{3+} needed for effective shift.	Part A: 0.0020 ± 0.0001 mol dm^{-3} to all stock solutions in different volumes using a micropipette. Part B: same volume as Fe^{3+} ($4.000 \pm 0.004 cm^3$)
Cuvette path length (l)	$A = \varepsilon cl$ shows varying l obstructs ε determination.	Same 1cm path length cuvette used throughout.
Temperature of stock solutions for Part A	Temperature fluctuation destabilizes ε and shifts equilibrium.	Measurements were handled at $19.7 \pm 0.1°C$.
Cuvette cleanliness	Fingerprints/droplets scatter light, which causes systematic error.	Wiped with Kimtech before every reading.
The depth of the thermometer inside the cuvette while reading its absorbance.	Probe too deep in the cuvette blocks the light path.	Held at ⅓ cuvette depth via clamp and stand.

Table 3: Table of controlled variables

3.7 Risk assessment

-Safety considerations[7]

Chemicals/Apparatus	Safety concerns	Prevention
Iron(III) nitrate	-Causes serious eye irritation	-Wear protective gloves/clothing/goggles

nonahydrate ($Fe(NO_3)_3 \cdot 9H_2O$)	-Causes skin irritation -May cause respiratory irritation -May intensify fire	-Avoid breathing dust/gas -Keep away from sparks, ignition sources
Potassium thiocyanate (KSCN)	-Harmful if inhaled/swallowed/in contact with skin -Harmful to aquatic life with long-lasting effects	-Keep the container tightly closed -Avoid releasing to the environment -Wear protective gloves/clothing/goggles
Nitric acid (HNO_3)	-Causes severe skin burns/eye damage -May intensify fire -May be corrosive to metals -Toxic if inhaled	-Wear protective gloves/clothing/goggles -Keep away from sparks/open flames/ ignition sources -Avoid breathing dust/fume/gas -Wash thoroughly after handling -Store in a corrosion-resistant container

Table 4: safety concerns

-Environmental/Ethical considerations

Environmental: The chemical species involved, especially SCN^-, are toxic to aquatic life as respiratory inhibitors. Furthermore, Fe^{3+} and $0.50_0 \pm 0.009$ mol dm^{-3} HNO_3 pose acidification risks and can corrode plumbing. Consequently, all waste was disposed of in designated chemical waste containers rather than laboratory drains.

Ethical: To ensure scientific integrity, all raw data were recorded and analyzed without any manipulation. Proper labelling and a clean workspace were maintained to prevent accidents that could harm peers.

3.8 Procedure

3.8.1 Part A: Creating a calibration curve for $[Fe(SCN)]^{2+}$

-Preparing solutions and calculating their exact molar concentration and absolute uncertainty

Mass of $Fe(NO_3)_3$ = 4.102 ± 0.001g (0.001g-digital scale x1) (4s.f.)
Volume of diluted $Fe(NO_3)_3$ solution = 0.0500 ± 0.0001dm^3 (50.0 ± 0.1cm^3-volumetric flask)(3s.f.)
Purity of $Fe(NO_3)_3$ = 98.5%

Molar mass of $Fe(NO_3)_3 \cdot 9H_2O$ = 404gmol^{-1}

Mass of KSCN = 0.020 ± 0.001g (0.001g-digital scale x1)(2s.f.)
Volume of diluted KSCN solution = 0.1000 ± 0.0002dm^3 (100.0 ± 0.2cm^3-volumetric flask)(4s.f.)
Purity of KSCN = 99%
Molar mass of KSCN = 97.18gmol^{-1}

Purity of HNO_3 = 65 ± 1% (2s.f.)
Density of HNO_3 = 1420gdm^{-3}
Molar mass of HNO_3 = 63.01gmol^{-1}
Volume of HNO_3 = 3.410 ± 0.004cm^3 (0.001cm^3-micropipette x4)(4s.f.)
Total volume of diluted HNO_3 = 100.0 ± 0.2 cm^3 (100.0 ± 0.2cm^3-volumetric flask x1)(4s.f.)

Type	Calculations	Error propagation
$0.200_0 \pm 0.0005$ mol dm^{-3} $Fe(NO_3)_3 \cdot 9H_2O$	$Moles\ (Fe^{3+}) = \frac{mass \times \frac{purity}{100}}{molar\ mass}$ $= \frac{4.102g \times \frac{98.5}{100}}{404gmol^{-1}} = 0.010001\ mol$ = 0.01000 (rounded to 4s.f.) $Molar\ concentration\ (Fe^{3+}) = \frac{Moles}{Volume}$ $= \frac{0.010001mol}{0.0500dm^3} = 0.200023moldm^{-3}$ = 0.200 (rounded to 3s.f.) = 0.200_0 (subscript to match the absolute uncertainty, but it is not a s.f.)	**Relative uncertainty of mass ($Fe(NO_3)_3 \cdot 9H_2O$)** = $\frac{0.001}{4.102}$ = 0.000244 **Relative uncertainty of volume ($Fe(NO_3)_3 \cdot 9H_2O$)** = $\frac{0.12}{50} = 0.0024$ **Error propagation** •Relative uncertainty (mol ($Fe(NO_3)_3 \cdot 9H_2O$)) = Relative uncertainty (mass + purity + molar mass) = 0.000244+0+0= 0.000244 •Relative uncertainty (Molar concentration (Fe^{3+}) = Relative uncertainty (moles + volume) = 0.000244+0.0024 = 0.002644 **Absolute uncertainty (molar concentration)** = $0.200023 \times 0.002644 = 0.000528818$ =0.0005 (1s.f.)

0.0020± 0.0001 mol dm^{-3} KSCN	$Moles\ (SCN^{-1}) = \frac{mass \times \frac{purity}{100}}{molar\ mass}$ $= \frac{0.020g \times \frac{99}{100}}{97.18gmol^{-1}} = 0.000204\ mol$ = 0.00020 (rounded to 2s.f.) $Molar\ concentration\ (SCN^{-1}) = \frac{Moles}{Volume}$ $= \frac{0.000204mol}{0.1000dm^3} = 0.002037moldm^{-}$ = 0.0020 (rounded to 2s.f.)	**Relative uncertainty of mass (KSCN)** = $\frac{0.001}{0.020} = 0.05$ **Relative uncertainty of volume (KSCN)** = $\frac{0.20}{100.0} = 0.0020$ **Error propagation** •Relative uncertainty (mol (KSCN)) = Relative uncertainty (mass + purity + molar mass) = 0.05+0+0= 0.05 •Relative uncertainty (Molar concentration (SCN^{-1}) = Relative uncertainty (moles + volume) = 0.05+0.0020) = 0.0520 **Absolute uncertainty (molar concentration)** = $0.002037 \times 0.0520 = 0.000105924$ =0.0001 (1s.f.)
0.50_0±0.009 mol dm^{-3} HNO_3	$Molar\ concentration\ (HNO_3) = density \times \frac{purity}{100} \div molar\ mass =$ $1420gdm^{-3} \times \frac{65}{100} \div 63.01gmol^{-1}$ =14.64847moldm^{-3} = 15(2s.f.) Molar concentration (diluted HNO_3) = $\frac{Concentration_{HNO_3} \times Volume_{HNO_3}}{Volume_{total}}$ $\frac{14.64847 \times 0.003410}{0.1000} = 0.499513$ moldm^{-3} = 0.50(rounded to 2s.f.) = 0.50_0 (subscript to match the absolute uncertainty, but it is not a	**Relative uncertainty of purity (HNO_3)** = $\frac{1}{65}$ = 0.015385 **Relative uncertainty of volume (HNO_3)** = $\frac{0.004}{3.410} = 0.001173$ **Relative uncertainty of volume (diluted HNO_3)** = $\frac{0.2}{100.0} = 0.0020$ **Error propagation** •Relative uncertainty of molar concentration (HNO_3) = relative uncertainty (density+purity+molar mass) = 0+0.015385+0 = 0.015385 •Relative uncertainty of molar concentration

	s.f.)	(diluted HNO_3) = relative uncertainty (molar concentration(HNO_3)+volume(HNO_3)+volume(diluted HNO_3) = 0.015385+0.001173+0.0020 = 0.018557636 **Absolute uncertainty of molar concentration (diluted HNO_3)** = $0.499513 \times 0.018557636 = 0.009269776$ = 0.009 (rounded to 1s.f.)
Molar concentration (Fe^{3+}) = $0.200_0 \pm 0.0005$ mol dm^{-3} Molar concentration (SCN^{-1}) = 0.0020 ± 0.0001 mol dm^{-3} Molar concentration (diluted HNO_3) = $0.50_0 \pm 0.009$ mol dm^{-3}		

Table 5: How the stock solutions were prepared

-Procedure for Part A

Stock solution(cm^3)	**std1**	**std2**	**std3**	**std4**	**std5**
$0.200_0 \pm 0.0005$ mol dm^{-3} $Fe(NO_3)_3 \cdot 9H_2O$	5.000(±0.005)	5.000(±0.005)	5.000(±0.005)	5.000(±0.005)	5.000(±0.005)
0.0020 ± 0.0001 mol dm^{-3} KSCN	0	0.500(±0.001)	1.000(±0.001)	1.500(±0.002)	2.000(±0.002)
$0.50_0 \pm 0.009$ mol dm^{-3} HNO_3	5.000(±0.005)	4.500(±0.005)	4.000(±0.004)	3.500(±0.004)	3.000(±0.003)

Table 6: Preparing standard solutions

1. 5 stock solutions filled by the beakers according to Table 6 were created using a micropipette(±0.001cm^3).
2. 5 cuvettes with 4.000±0.004cm^3 of each stock solution were filled using a micropipette.
3. 1 cuvette with 4.000±0.004cm^3 0.50_0±0.009 moldm^{-3} HNO_3 was filled.
4. The HNO_3 cuvette was used as the blank for the spectrophotometer
5. The absorbance of each standard solution was measured 5 times.
6. The values of absorbance and its standard deviation were averaged.

3.8.2 Part B: Effect of temperature on the equilibrium constant of $[Fe(SCN)]^{2+}$.

-Preparing 0.00200_0±0.000009 mol dm^{-3} Fe^{3+} and calculating their exact molar concentration and absolute uncertainty

Molar concentration of Fe^{3+} stock = 0.200_0 ± 0.0005moldm^{-3} (3s.f.)
Volume of Fe^{3+} stock = 1.000 ± 0.001 cm^3 (0.001cm^3-micropipette x1)(4s.f.)
Volume of diluted Fe^{3+} solution = 100.0 ± 0.1cm^3 (100.0 ± 0.1cm^3-volumetric flask)(4s.f.)

Calculations	**Error propagation**
Molar concentration (diluted Fe^{3+}) = $\frac{Concentration_{Fe^{3+}stock} \times Volume_{Fe^{3+}stock}}{Volume_{diluted\ Fe^{3+}}}$ $= \frac{0.200023 \times 0.001000}{0.1000} = 0.00200$ mol dm^{-3} (rounded to 3s.f.) = 0.00200_0 (subscript to match the absolute uncertainty, but it is not a s.f.)	**Relative uncertainty of molar concentration (Fe^{3+} stock)** = $\frac{0.000528818}{0.200023} = 0.002644$ **Relative uncertainty of volume (Fe^{3+} stock)** = $\frac{0.001}{1.000}$ = 0.001 **Relative uncertainty of volume (diluted Fe^{3+})** = $\frac{0.1}{100.0} = 0.001$ **Error propagation** •Relative uncertainty of molar concentration (diluted Fe^{3+}) = relative uncertainty (molar concentration (Fe^{3+} stock)+volume (Fe^{3+} stock)+volume (diluted Fe^{3+})) = 0.002644+0.001+0.001 = 0.004644 **Absolute uncertainty of molar concentration (diluted Fe^{3+})** = 0.00200×0.004644 = 0.000009289 = 0.000009 (rounded to 1s.f.)

Molar concentration (diluted Fe^{3+}) = $0.00200_0 \pm 0.000009$ mol dm^{-3}

Table 7: Preparing the diluted iron solution

-Procedure for Part B



Figure1: Image to show how the thermometer was placed for a constant temperature reading.

1. A sample was made by mixing 4.000cm^3 (±0.004) of 0.00200_0±0.000009$moldm^{-3}$ $Fe(NO_3)_3 \cdot 9H_2O$, 2.000cm^3 (±0.002) 0.50_0±0.009$moldm^{-3}$ HNO_3, and 4.000cm^3 (±0.004) 0.0020±0.0001$moldm^{-3}$ KSCN in a 50cm^3 beaker.
2. 4.000cm^3 (±0.004) of the sample was put into a cuvette using a micropipette.
3. 0.50_0±0.009$moldm^{-3}$ HNO_3 was used as the blank for the spectrophotometer.
4. Measuring the target temperature for 20.0±0.1°C.
    a. The absorbance was measured at 19.7±0.1°C 3 times.
5. Measuring the target temperature for 50.0±0.1°C, 40.0±0.1°C, and 30.0±0.1°C.
    a. The sample cuvette was placed into 50cm^3 beaker filled with distilled water on a hot plate (Do not let the distilled water enter the cuvette).
    b. The sample was heated to 55.4±0.1°C, measured with a thermometer (±0.1°C).
    c. The cuvette was retrieved, and the wet surface was wiped with Kimtech, then quickly placed into the spectrophotometer (The temperature was constantly measured as shown in Figure1).
    d. The absorbance of the sample was measured as it reached 50.0±0.1°C, 40.0±0.1°C, and 30.0±0.1°C, and the temperature was recorded when the button was pressed and when the measurement was complete.
    e. The trial was repeated 2 more times.

6. Measuring the target temperature 10.0±0.1°C.
    a. The cuvette containing the sample was placed in a beaker of ice water until the temperature reached 4.1±0.1°C.
    b. The cuvette was retrieved, and the wet surface was wiped with Kimtech, then quickly placed in the spectrophotometer (The temperature was constantly measured as shown in Figure1).
    c. The absorbance of the sample was measured as it reached 10.0±0.1°C, and the temperature was recorded when the button was pressed and when the measurement was complete.
    d. The trial was repeated 2 more times.

IV. Data

4.1 Part A: Calibration

4.1.1 Qualitative data

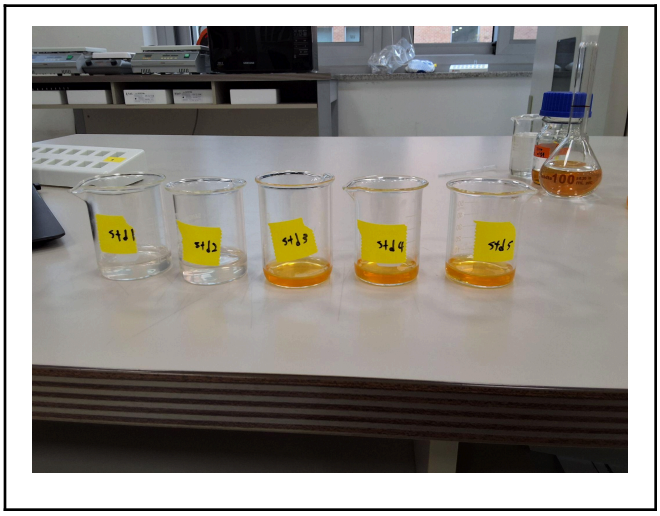


Figure2: The two solutions on the left are $Fe(NO_3)_3 \cdot 9H_2O$ mixed with HNO_3, and the rest are before the addition of acid

The $0.200_0 \pm 0.0005 moldm^{-3}$ Fe^{3+} stock was vivid orange, but turned colourless with slight fogging once mixed with $0.50_0 \pm 0.009 moldm^{-3}$ of HNO_3, showing acid is essential to control Fe^{3+} hydrolysis for accurate absorbance readings.

4.1.2 Quantitative data

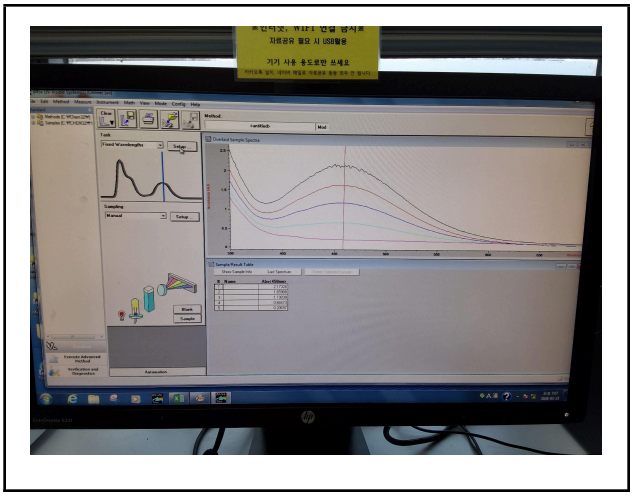


Figure3: The absorbance curve for the standard solutions, and its peak (458nm)

The absorbance gained from the UV was rounded to the 4th decimal place due to the standard deviation.

-Raw data

Volume of 0.200_0±0.0005mold m^{-3} $Fe(NO_3)_3 \cdot 9H_2O$ (±0.005c m^3)	**Volume of 0.0020±0.0001mold m^{-3} KSCN (cm^3)**	**Volume of 0.50_0±0.009moldm^{-3} HNO_3**	**Absorbance (±standard deviation)**				
			Trial1	**Trial2**	**Trial3**	**Trial4**	**Trial5**
5.000	0	5.000 (±0.005)	0.2070 (±0.0004)	0.2061 (±0.0004)	0.2017 (±0.0005)	0.1990 (±0.0002)	0.2040 (±0.0005)
5.000	0.500 (±0.001)	4.500 (±0.005)	0.6673 (±0.0003)	0.6572 (±0.0004)	0.6431 (±0.0003)	0.6483 (±0.0004)	0.6378 (±0.0004)
5.000	1.000 (±0.001)	4.000 (±0.004)	1.1903 (±0.0004)	1.1894 (±0.0003)	1.1795 (±0.0004)	1.1780 (±0.0003)	1.1715 (±0.0003)

5.000	1.500 (±0.002)	3.500 (±0.004)	1.6590 (±0.0004)	1.6668 (±0.0004)	1.6478 (±0.0004)	1.6499 (±0.0004)	1.6418 (±0.0004)
5.000	2.000 (±0.002)	3.000 (±0.003)	2.1732 (±0.0003)	2.1750 (±0.0003)	2.1855 (±0.0004)	2.1801 (±0.0003)	2.1805 (±0.0004)

Table 8: Raw data of the absorbance of known concentration

Since the amount of Fe^{3+} present in each solution is more than 100 times that of SCN^-, it is assumed that all of SCN^- reacted into $[Fe(SCN)]^{2+}$ according to Le Chatelier's principle. Therefore, the concentration of SCN^- after mixing is the same as the concentration of $[Fe(SCN)]^{2+}$ formed.

-Sample calculation: Concentration of $SCN^-/[Fe(SCN)]^{2+}$ moldm^{-3} for 1.000±0.001cm^3 of SCN^- from Table 8

Molar concentration of SCN^- stock = 0.0020 ± 0.0001 moldm^{-3}(2s.f.)
Volume of SCN^- stock = 1.000 ± 0.001cm^3 (0.001cm^3-micropipette x1)(4s.f.)
Total volume of standard solution = 10.00 ± 0.01cm^3 (0.001cm^3-micropipette x10)(4s.f.)

Calculations	**Error propagation**
Molar concentration ($SCN^-/[Fe(SCN)]^{2+}$) = $\frac{Concentration_{SCN^-} \times Volume_{SCN^{-1}}}{Volume_{total}}$ = $\frac{0.002037 moldm^{-3} \times 0.001000 dm^3}{0.01000 dm^3}$ = 0.00020375 = 0.00020(rounded to 2s.f.)	**Relative uncertainty of Molar concentration (SCN^{-1} stock)** = $\frac{0.000105924}{0.002037}$ = 0.0520 **Relative uncertainty of volume (SCN^{-1})** = $\frac{0.001}{1.000}$ = 0.001 = 0.001 **Relative uncertainty of volume (total)** = $\frac{0.01}{10.00}$ = 0.001 **Error propagation** Relative uncertainty of Molar concentration (SCN^-) = relative uncertainty (SCN^- stock+volume(SCN^- stock)+total volume) = 0.0520+0.001+0.001 = 0.0540 **Absolute uncertainty of Molar concentration ($SCN^-/[Fe(SCN)]^{2+}$)** = 0.00020375×0.0540 =

	0.0000110025 = 0.00001(rounded to 1s.f.)
Molar concentration ($[Fe(SCN)]^{2+}$) = 0.00020 ± 0.00001 $moldm^{-3}$	

Table 9: Sample calculation of the concentration of complex ions in standard solutions

-Processed data

Concentration of [FeSCN]2+ formed ($moldm^{-3}$) ($x10^{-4}$)	**Average absorbance** (±average standard deviation) **at 458nm**
0	0.2035 (±0.0004)
1.0 (±0.1)	0.6507 (±0.0004)
2.0 (±0.1)	1.1818 (±0.0004)
3.1 (±0.2)	1.6531 (±0.0004)
4.1 (±0.2)	2.1789 (±0.0003)

Table 10: Processed data of complex ion concentration and their absorbance

-Graph

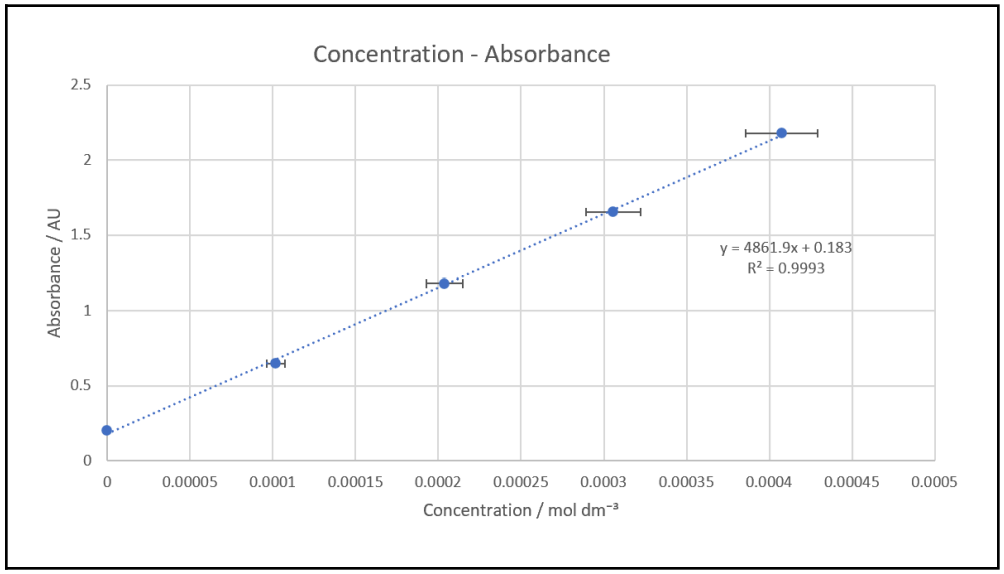


Figure 4: The graph of average absorbance (AU) plotted against concentration (moldm^-3)

The Beer-Lambert Law requires a y-intercept of zero. However, the calibration curve from Figure4 showed a non-zero intercept of 0.183, indicating that the absorbance of Fe^{3+} is not negligible despite the addition of HNO_3.

To fix this, the average absorbance of the 0 $moldm^{-3}$ (0.2035±0.0004) was subtracted from all data points, and the intercept was set to 0. Due to the substantial excess of Fe^{3+}, it was assumed that the concentration of unreacted iron remained constant across all standard solutions. Consequently, the absolute uncertainty of the corrected absorbance was increased by 0.0004 due to the subtraction.

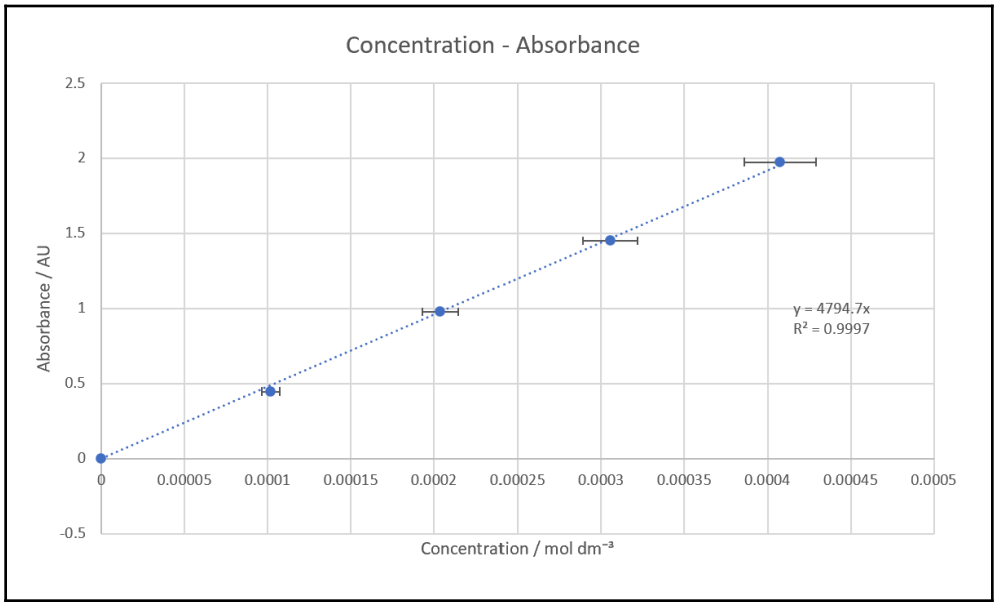


Figure5: Refined graph

The vertical error bars are barely visible despite being marked on the graph. It is possible to use the maximum/minimum gradients and to obtain the uncertainty of the gradient.

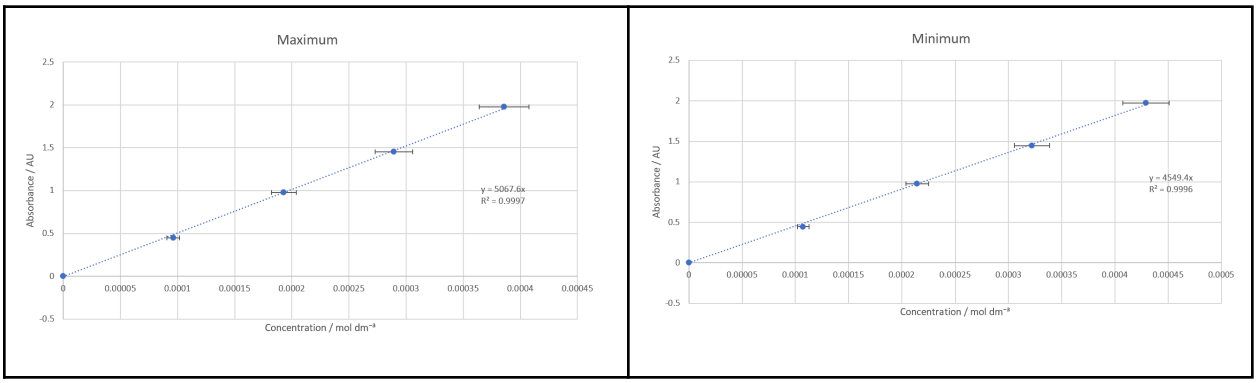


Figure 6: The maximum/minimum gradients of the graph

The maximum gradient was derived by plotting (0, 0) through data points $(c - \Delta c,\ A + \Delta A)$, while the minimum gradient used $(c + \Delta c,\ A - \Delta A)$ for all data points.

A larger difference between the gradient and its maximum/minimum was used as the absolute uncertainty.
5067.6-4794.7 = 272.9 = 270 (rounded to 2s.f.)
4794.7-4549.4 = 245.3 = 250 (rounded to 2s.f.)
Therefore, the molar extinction coefficient is 4790±270.

4.2 Part B: The equilibrium constant as temperature varies

4.2.1 Qualitative data

Bubbles formed inside the cuvette as the temperature increased, indicating a systematic error and the limitation of my experiment.

4.2.2 Quantitative data

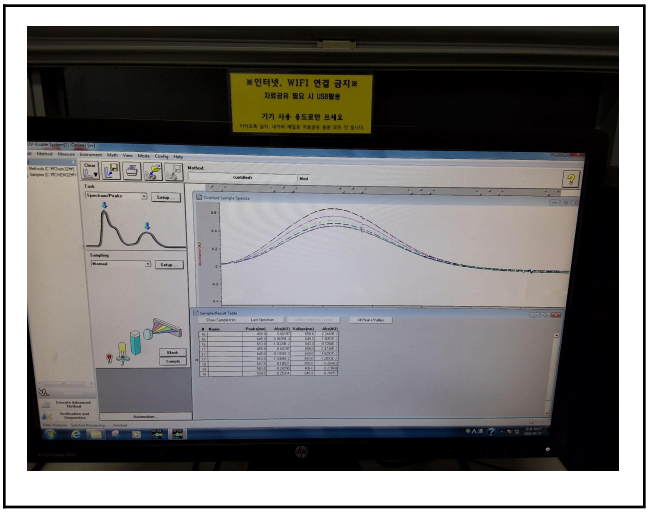


Figure7: The absorbance curve for the sample at different temperatures (top curve being at 11.2±0.1°C and bottom curve at 49.8±0.1°C)

-Raw data

Target temperature	Trial	Initial temperature (±0.1°C)	Final temperature (±0.1°C)	Absorbance
50.0±0.1°C	1	50.1	50.0	0.4724 (±0.0004)
	2	50.3	49.7	0.4699 (±0.0003)

	3	49.7	49.1	0.4723 (±0.0005)
40.0±0.1°C	1	40.4	40.0	0.5002 (±0.0003)
	2	40.0	39.6	0.5010 (±0.0003)
	3	39.6	39.2	0.5021 (±0.0003)
30.0±0.1°C	1	30.4	30.3	0.5333 (±0.0003)
	2	30.3	30.2	0.5338 (±0.0003)
	3	30.0	29.9	0.5339 (±0.0004)
20.0±0.1°C	1	19.7	19.7	0.5825 (±0.0004)
	2	19.7	19.7	0.5829 (±0.0003)
	3	19.7	19.7	0.5824 (±0.0003)
10.0±0.1°C	1	10.7	10.9	0.6656 (±0.0004)
	2	11.5	11.6	0.6620 (±0.0004)
	3	11.1	11.1	0.6630 (±0.0004)

Table 11: Raw data of the absorbance of the sample at different temperatures

-Processed data

Average temperature (±0.1°C)	**Average absorbance** (±average standard deviation)
49.8	0.4715 (±0.0004)
39.8	0.5011 (±0.0003)
30.2	0.5336 (±0.0004)
19.7	0.5826 (±0.0003)

11.2	0.6635 (±0.0004)

Table 12: Processed data by averaging raw data from Table 11

We can divide its absorbance by the molar extinction coefficient (4790±270) to obtain the concentration of $[Fe(SCN)]^{2+}$ for the corresponding temperatures.

-Sample calculation: Calculating the concentration of $[Fe(SCN)]^{2+}$(For sample at 30.2±0.1°C)

Molar extinction coefficient = 4790 ± 270 $dm^3mol^{-1}cm^{-1}$(3s.f.) Absorbance: 0.5336 ± 0.0004 (4AU-UV-vis spectrophotometer)(4s.f.) Path length: 1cm	
Calculations	**Error propagation**
Molar concentration ($moldm^{-3}$) = $\frac{Absorbance}{Molar\ extinction\ coefficient(dm3mol^{-1}cm^{-1})\times Path\ length(cm)}$ = $\frac{0.533641333}{4794.7\times1} = 0.000111298 = 0.000111$ (rounded to 3s.f.)	**Relative uncertainty of Molar extinction coefficient** = $\frac{272.9}{4794.7}$ = 0.056917 **Relative uncertainty of Absorbance** = $\frac{0.00036046}{0.533641333}$ = 0.000675 **Error propagation** •Relative uncertainty of molar concentration ($[Fe(SCN)]^{2+}$ = Relative uncertainty (Molar extinction coefficient+absorbance) = 0.056917+0.000675 = 0.057592 **Absolute uncertainty of molar concentration ($[Fe(SCN)]^{2+}$** = 0.000111298×0.057592 = 0.000006410 = 0.000006 (1s.f.)
Molar concentration ($[Fe(SCN)]^{2+}$ = (1.11 ± 0.06)×10^{-4} $moldm^{-3}$	

Table 13: Sample calculation of the concentration of complex ions at different temperatures

-Processed data

Average temperature (±0.1°C)	**Concentration (±absolute uncertainty) ($x10^{-4}$)**
49.8	0.98 (±0.06)
39.8	1.05 (±0.06)
30.2	1.11 (±0.06)
19.7	1.22 (±0.07)
11.2	1.38 (±0.08)

Table 14: Processed data of complex ions' concentration at different temperatures

The concentration of reactant species at equilibrium can be obtained by stoichiometry. From the reaction formula, we know that the molar ratio between all species is 1:1. With the initial concentration of reactants and the final concentration of products, we can calculate the equilibrium constant.

According to the distribution of the sample solution, we can calculate the initial concentration of reactants.

	0.00200_0 ± $0.000009 moldm^{-3}$ Fe^{3+}	**$0.0020 \pm 0.0001 moldm^{-3}$ SCN^-**	**0.50_0 ±$0.009 moldm^{-3}$ HNO_3**
Volume added (cm^3)	4.000 (±0.004)	4.000 (±0.004)	2.000 (±0.002)

Table 15: The distribution of stock solutions in the sample

-Sample calculation: Initial concentration calculations using $0.0020 \pm 0.0001 moldm^{-3}$ SCN^-

Concentration of SCN^- stock: $0.0020 moldm^{-3} \pm 0.0001 moldm^{-3}$(2s.f.) Volume of SCN^- solution added: $4.000 cm^3$ ±0.004 cm^3 ($0.001 cm^3$-micropipette x4)(4s.f.) Total volume: $10.00 cm^3$ ± 0.01 ($0.001 cm^3$-micropipette x10)(4s.f.)	
Calculations	**Error propagation**

$\text{Concentration}_{\text{Initial}} = \frac{Concentration_{SCN^-} \times Volume_{SCN^-}}{Volume_{total}}$ $= \frac{0.00203746 \times 0.004000}{0.01000} = 0.00081498 =$ 0.00081(2s.f.)	**Relative uncertainty of molar concentration (SCN^- stock)** = $\frac{0.000105924}{0.00203746} = 0.0520$ **Relative uncertainty of volume (SCN^- added)** = $\frac{0.004}{4.000} = 0.001$ **Relative uncertainty of volume (total)** = $\frac{0.01}{10.00}$ = 0.001 **Error propagation** •Relative uncertainty of Molar concentration (initial) = Relative uncertainty (molar concentration (SCN^- stock)+volume (SCN^- added)+Relative uncertainty of volume (total)) = 0.0520+0.001+0.001 = 0.0540 **Absolute uncertainty of Molar concentration (initial)** = 0.00081498×0.0540 = 0.000044009 = 0.00004 (rounded to 1s.f.)
Concentration of initial SCN^{-1}: 0.00081 ± 0.00004 $moldm^{-3}$	

Table 16: Sample calculation of the initial concentration of SCN^- in the sample

The concentration attained from absorbance was 0.000111, so the change in concentration is 0.000111 for all species.

-Equilibrium concentrations of species at 30.2±0.1°C

	$[Fe^{3+}]$	$[SCN^-]$	$[[Fe(SCN)]^{2+}]$
Initial Concentration ($moldm^{-3}$)	0.000800 ± 0.000005	0.00081 ± 0.00004	0
Change in concentration	-(0.000111 ± 0.000006)	-(0.000111 ± 0.000006)	+(0.000111 ± 0.000006)

($moldm^{-3}$)			
Concentration at equilibrium	0.000689±0.000011	0.00070±0.00005	0.000111 ± 0.000006

Table 17: Table of concentration of each reagent at equilibrium

The concentration at equilibrium was substituted into the equation for the equilibrium constant,

$$K_C = \frac{[[Fe(SCN)]^{2+}]_{eq}}{[Fe^{3+}]_{eq}[SCN^{-}]_{eq}}$$

-Sample calculation: Equilibrium constant at 30.2±0.1°C

Molar concentration at equilibrium (Fe^{3+}): 0.000689 ± 0.000011 (3s.f.)

Molar concentration at equilibrium (SCN^{-}): 0.00070 ± 0.00005 (2s.f.)

Molar concentration at equilibrium ($[Fe(SCN)]^{2+}$): 0.000111 ± 0.000006 (3s.f.)

Calculations	**Error propagation**
$K_C = \frac{[[Fe(SCN)]^{2+}]_{eq}}{[Fe^{3+}]_{eq}[SCN^{-}]_{eq}}$ $= \frac{0.000111298}{0.000688795 \times 0.000703684} = 229.6255526 = 230$ (rounded to 2s.f.)	**Relative uncertainty of Molar concentration (Fe^{3+})** $= \frac{0.000011726}{0.000688795} = 0.017023331$ **Relative uncertainty of Molar concentration (SCN^{-})** $= \frac{0.000050419}{0.000703684} = 0.071649945$ **Relative uncertainty of Molar concentration ($[Fe(SCN)]^{2+}$)** $= \frac{0.000006410}{0.000111298} = 0.057592485$ **Error propagation** Relative uncertainty of K_C = Relative uncertainty (Molar concentration(Fe^{3+})+Molar concentration(SCN^{-})+Molar concentration($[Fe(SCN)]^{2+}$)) = 0.017023331+0.071649945+0.057592485 = 0.146266

	Absolute uncertainty of K_C = 229.6255526× 0.146266 = 33.58636 = 30 (rounded to 1s.f.)
Equilibrium constant = 230 ± 30	

Table 18: Sample calculation of equilibrium constant for the sample at 30.2±0.1°C

Therefore, the equilibrium constants at different temperatures are:

-Processed data

Temperature of the sample (±0.1K)	**Value of K_C with its absolute uncertainty**
323.0	200 ± 30
313.0	210 ± 30
303.3	230 ± 30
292.9	260 ± 40
284.3	310 ± 50

Table 19: Processed data of equilibrium constants at different temperatures

-Graph

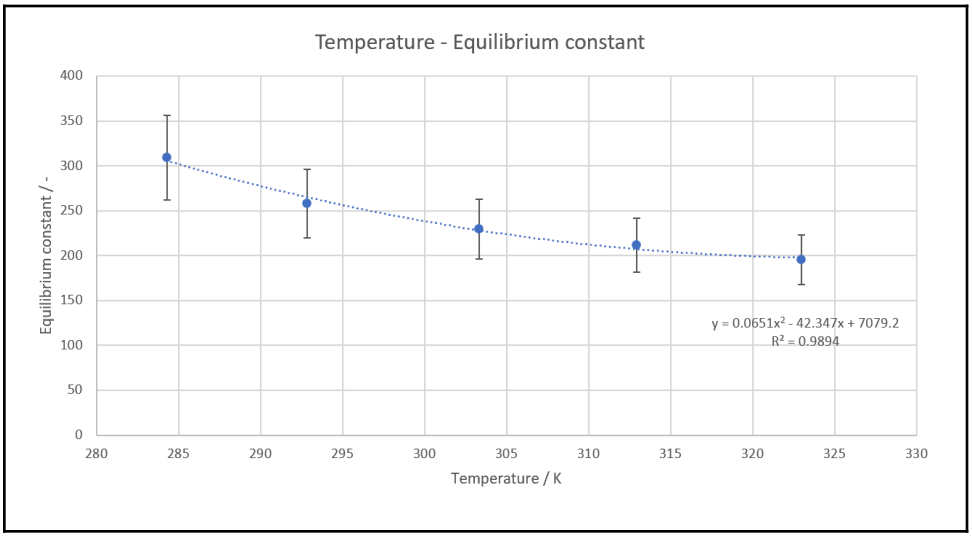


Figure 8: The relationship between temperature and equilibrium constant plotted in a graph

V. Conclusion/Evaluation

The primary objective of this investigation was to determine the relationship between temperature (10.0±0.1°C, 20.0±0.1°C, 30.0±0.1°C, 40.0±0.1°C, 50.0±0.1°C) and the equilibrium constant for the formation of $[Fe(SCN)]^{2+}$ complex. As evidenced by the processed data in Table19 and the corresponding plot in Figure8, the results align with the initial hypothesis: the equilibrium constant decreases as temperature increases. Specifically, K_C decreased from 310±50 at 11.2±0.1°C to 200±30 at 49.8±0.1°C. The observed trend can be explained through Le Chatelier's principle. Since the formation of $[Fe(SCN)]^{2+}$ is an exothermic process ($\Delta H^\circ < 0$), heat is treated as a product of the reaction:

$$Fe^{3+}_{(aq)} + SCN^-_{(aq)} \rightleftharpoons [Fe(SCN)]^{2+}_{(aq)} + \text{heat}$$

The system tries to counteract the addition of thermal energy by shifting the equilibrium position in the endothermic direction (reverse reaction) to absorb the excess heat. This shift results in a lower equilibrium concentration of $[Fe(SCN)]^{2+}$, resulting in a decrease in the equilibrium constant. However, Figure8 shows the trend approaching an asymptote as temperature increases. This behavior is validated by the van't Hoff equation, which relates the equilibrium constant to temperature:

$$lnK_C = -\frac{\Delta H^\circ}{R}\left(\frac{1}{T}\right) + \frac{\Delta S^\circ}{R}$$

By plotting the natural logarithm of the equilibrium constant against the inverse of temperature, we get a linear graph:

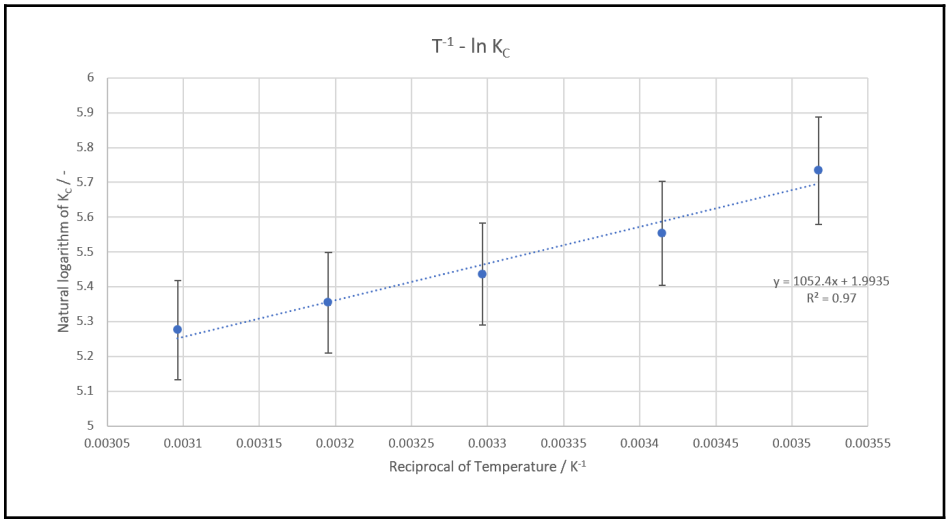


Figure9: Natural logarithm of equilibrium constant plotted against inverse of temperature (K^{-1})

For an exothermic reaction, where ΔH° is negative, the term $-\frac{\Delta H^{\circ}}{R}$ is positive. Consequently, as the absolute temperature (T) decreases, the value $\frac{1}{T}$ increases, leading to an enlargement in $\ln K_C$ and thus an increase in K_C.

The solid observation was possible due to these factors:

Strengths	Explanation
Repeated measurements	Reduced random error, leading to no significant outliers.
Micropipette ($\pm\ 0.001 cm^3$)	Minimized propagated uncertainty from error propagation.
UV-vis spectrophotometer	Precise indirect concentration measurement with a significantly small relative uncertainty.
HNO_3 addition	Reversed Fe^{3+} hydrolysis, avoiding a major absorbance error seen in preliminary trials.

Table20: Strengths and explanations

In conclusion, the experiment was able to successfully answer the research question. The data links to well-established principles, which strongly explain the trend. Calculating the %error for the standard enthalpy change for the trend (gradient) would be ideal to understand the deviation from the theoretical value for the relationship. However, the literature value could not be found. Thus, the %error of the precise equilibrium constant at 20°C (the only literature value I could find) was used (K_C=113).[8]

$$\%\text{error} = \frac{|258.205-113|}{113} \times 100(\%) = 128.5\% = 130\% \text{ (rounded to 2s.f.)}$$

Such a high %error indicates that, despite the overall trend following the hypothesis, the actual values are different. Since 130% is far above the % relative uncertainty (15%), this must be a systematic error. To understand the potential cause, the limitations were listed below alongside improvements.

Limitations	Explanation	Improvements
Ionic strength differs from the literature	Other ions, mainly H^+ and NO_3^- from HNO_3, surround Fe^{3+} and SCN^- and partly block their charges, so fewer complexes form, and K_C falls as ionic strength rises. K_C is 710 with no other ions[9], 146 at 0.40 and 114 at 1.28moldm^{-3} (25°C)[10]. Mine was about 0.1moldm^{-3}, so 258 is reasonable.	Make Part B solutions in 0.4moldm^{-3} HNO_3 similar to the literature conditions to be fair when comparing the values.
Not all SCN^- reacted in the standards	Part A assumed all SCN^- formed the complex, but some always stays free at equilibrium. Rearranging K_C gives $[Fe(SCN)]^{2+} \div [\text{free } SCN^-] = K_C \times [Fe^{3+}] = 26$, so about 1 in 27 SCN^- (4%) stayed free. The standards had less complex than calculated, so ε was smaller. In Part B, A÷ε then overestimated $[Fe(SCN)]^{2+}$, so every K_C is at least 5% higher.	Remake std5 with 0.025, 0.050 and 0.100moldm^{-3} Fe^{3+}. If A rises with more Fe^{3+}, some SCN^- was unreacted. A plot of $\frac{1}{A}$ against $\frac{1}{[Fe^{3+}]}$ is a straight line, and its y-intercept (unlimited Fe^{3+}) gives $\frac{1}{A}$ for complete reaction. The true ε is then A÷[SCN^-].
Hydrated KSCN/$Fe(NO_3)_3 \cdot 9H_2O$ (10-year-old reagents)	Both solids absorb water from the air, so part of the weighed mass was water, not reagent. The real moles, and so the real concentrations, were therefore lower than what was calculated. As reagents are in the denominator, it would have made the K_C smaller, so it doesn't explain my high K_C, but it is worth mentioning.	Dehydrate KSCN by heating to constant mass (100-173°C). For $Fe(NO_3)_3 \cdot 9H_2O$ (mp 47.2°C, can't be heated the same way), use redox titration for exact molarity or use fresh reagents.
ε only measured at 19.7°C	ε is the gradient of a calibration line made only at 19.7°C, but I used it at every temperature. Temperature changes ε, so this could change the trend.	Plot a calibration line at 10.0, 30.0, 40.0, and 50.0°C and use each gradient as ε.
Gas produced	Heating releases dissolved gas as bubbles,	Degas distilled water by boiling, then

while heating	which scatter light and raise absorbance at all wavelengths. However, Figure7 curves overlap at long wavelengths, where complex barely absorbs, so the effect was small.	cool in a sealed container before using in Part B.

Table21: Limitations, Explanations, and Improvements

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